

**Learning Assessment for Master of Engineering/ Master of Science in
Transportation Engineering (2024)**
October 2025

This program aims to achieve the following goals for student learning:

- **Learning Goal 1 (G₁):** Demonstrating learning of fundamental knowledge of transportation engineering with emphases on transportation system planning, data analytics, evaluation, and performance-based decision-making in support of sustainable transportation development.
- **Learning Goal 2 (G₂):** Demonstrating the ability to apply fundamental knowledge of transportation engineering to real-world transportation problems.

Performance Measures: Multiple performance measures are used for each learning goal to assess the extent to which the learning goal is achieved.

- Performance measures (PM) under Goal 1
 - PM₁₁: Clear comprehension and explanation of questions
 - PM₁₂: Format and completeness of answers provided
 - PM₁₃: Technical accuracy
 - PM₁₄: Ethical issues: Free of plagiarism and proper citing of existing work
- Performance measures under Goal 2
 - PM₂₁: Clear comprehension of the application problem
 - PM₂₂: Format and completeness of solution: Problem definition, literature review, proposed methodology, application, discussion of findings, and conclusion
 - PM₂₃: Depth of investigation
 - PM₂₄: Ethical issues: Free of plagiarism and proper citing of existing work.

Performance Targets: Table 1 shows relative weights assigned to learning goals and performance measures, as well as preset performance target of student learning.

Table 1. Relative weights of learning goals and performance measures, and performance target

Relative weights	Goal 1 α ($0.65 \leq \alpha \leq 0.8$)				Goal 2 $1 - \alpha$				Overall
	PM ₁₁	PM ₁₂	PM ₁₃	PM ₁₄	PM ₂₁	PM ₂₂	PM ₂₃	PM ₂₄	
	0.20	0.15	0.40	0.25	0.20	0.15	0.40	0.25	
Total points	100	100	100	100	100	100	100	100	100
Performance target	70%	70%	70%	100%	70%	70%	70%	100%	
Target performance points	14α	10.5α	28α	25α	$14(1-\alpha)$	$10.5(1-\alpha)$	$28(1-\alpha)$	$25(1-\alpha)$	77.5

Data Sources for Assessment: Data records of two graduate-level courses offered for transportation engineering program in calendar year 2024 are used for student learning assessment. Table 2 presents a summary of data sources and rubrics used for data analysis.

- List of courses, semesters, and sample sizes in which artifacts are collected
 - CAE544- Urban transportation planning (Core course): Fall 2024 (11 students)
 - CAE568- Transportation asset management (Elective): Spring 2024 (7 students).

Table 2. Summary of data sources

Course	Sample Size	Subject Area	Learning Goal	Evaluation Component					Total
				Homework	Term Paper	Case Studies/Reading Assignments	Midterm	Final	
CAE544: Urban Transportation Planning	Fall 2018 (11)	Distribution of total points		25%	15%	15%	20%	25%	100%
		a. Demand forecasting	G ₁	HW1, 2, 3			P1-10	P1-4	
			G ₂		TP/2	RA1-5			
		b. Demand/supply equilibrium	G ₁	HW4				P5-10	
CAE568: Transportation Asset Management	Spring 2018 (7)		G ₂		TP/2				100%
		Distribution of total points		30%	20%		20%	30%	
		a. Data sampling, collecting, processing, and database management	G ₁	HW1			P1-5	P3,4,7, (P11)/4	
			G ₂		TP/4				
		b. Transportation system performance modeling	G ₁	HW2			P6,7	P5,8, (P11)/4	
			G ₂		TP/4				
		c. Physical facility management	G ₁	HW3,4			P8,9	(P1,2)/2, P9, (P11)/4	
			G ₂		TP/4				
		d. System usage management	G ₁	HW5			P10-12	(P1,2)/2, P6,10, (P11)/4	
			G ₂		TP/4				

Note: G- Student learning goal; HW- Homework assignment; TP- Term paper; CS- Case study; and P- Midterm or final exam problem.

- Reporting of data results: Summary statistics and boxplot parameters characterizing student learning data distributions in conjunction with performance targets will be reported by subject area and by performance measure under each learning goal.
- Month/year of assessment: End of Spring and Fall semesters of calendar year 2024.
- Names/titles of evaluators: Academic Program Coordinator and Department Chair.

Results: Tables 3 and 4 exhibit student learning assessment results.

Table 3. Summary statistics and boxplot parameters of student learning data distributions by subject area and learning goal

CAE544 (11 students)	a. Demand forecasting		b. D/S equilibrium						Goal 1	Goal 2	Overall
	Goal 1	Goal 2	Goal 1	Goal 2							
Total points	40.5	22.5	29.5	7.5					70.0	30.0	100.0
Target points	31.4	17.4	22.9	5.8					54.3	23.3	77.5
Mean	34.0	19.1	24.5	6.5					58.4	25.6	84.0
STDEV	0.8	0.5	1.0	0.2					1.7	0.7	2.1
Min	32.4	18.0	22.8	6.0					55.2	24.0	80.9
Max	35.0	19.9	25.5	6.8					60.5	26.6	86.7
25 th percentile	33.7	18.8	23.9	6.4					56.9	25.4	82.0
Median	34.2	19.1	24.4	6.5					58.2	25.5	84.3
75 th percentile	34.7	19.3	25.5	6.8					60.1	26.0	85.5
Boxplot LB whisker	30.9	17.2	20.3	5.4					50.4	23.0	75.6
Boxplot UB whisker	36.2	20.1	28.0	7.3					64.8	27.1	90.8

CAE568 (7 students)	a. Data issues		b. Performance modeling		c. Facility mgmt		d. System usage mgmt		Goal 1	Goal 2	Overall
	Goal 1	Goal 2	Goal 1	Goal 2	Goal 1	Goal 2	Goal 1	Goal 2			
Total points	20.5	5.0	15.2	5.0	22.1	5.0	22.3	5.0	80.0	20.0	100.0
Target points	15.9	3.9	11.8	3.9	17.1	3.9	17.2	3.9	62.0	15.5	77.5
Mean	17.9	4.3	12.9	4.3	18.8	4.3	18.9	4.3	68.5	17.1	85.6
STDEV	0.7	0.1	0.5	0.1	0.6	0.1	0.7	0.1	2.1	0.5	2.5
Min	17.1	4.1	12.2	4.1	17.9	4.1	18.0	4.1	66.5	16.4	83.2
Max	18.7	4.5	13.4	4.5	19.5	4.5	20.1	4.5	71.7	18.0	89.7
25 th percentile	17.3	4.2	12.3	4.2	18.4	4.2	18.5	4.2	66.6	16.8	83.6
Median	17.7	4.3	13.0	4.3	18.5	4.3	18.8	4.3	67.3	17.2	84.2
75 th percentile	18.5	4.3	13.3	4.3	19.4	4.3	19.6	4.3	70.5	17.4	87.8
Boxplot LB whisker	15.3	3.9	10.8	3.9	16.4	3.9	16.3	3.9	60.6	15.5	76.9
Boxplot UB whisker	20.3	4.6	14.8	4.6	20.9	4.6	21.3	4.6	76.3	18.3	94.1

Table 4. Summary statistics and boxplot parameters of student learning data distributions by performance measure under each learning goal

CAE544 (11 students)	Goal 1				Goal 2				Overall
	PM ₁₁	PM ₁₂	PM ₁₃	PM ₁₄	PM ₂₁	PM ₂₂	PM ₂₃	PM ₂₄	
Target points	11.0	8.3	22.1	12.8	4.8	3.6	9.6	5.3	77.5
Mean	11.7	8.8	23.4	14.6	5.1	3.8	10.2	6.4	84.0
STDEV	0.3	0.3	0.7	0.4	0.1	0.1	0.3	0.2	2.1
Min	11.0	8.3	22.1	13.8	4.8	3.6	9.6	6.0	80.9
Max	12.1	9.1	24.2	15.1	5.3	4.0	10.6	6.7	86.7
25 th percentile	11.4	8.5	22.8	14.2	5.1	3.8	10.1	6.3	82.0
Median	11.6	8.7	23.3	14.6	5.1	3.8	10.2	6.4	84.3
75 th percentile	12.0	9.0	24.0	15.0	5.2	3.9	10.4	6.5	85.5
Boxplot LB whisker	10.1	7.6	20.2	12.6	4.6	3.4	9.2	5.7	75.6
Boxplot UB whisker	13.0	9.7	25.9	16.2	5.4	4.1	10.8	6.8	90.8

CAE568 (7 students)	Goal 1				Goal 2				Overall
	PM ₁₁	PM ₁₂	PM ₁₃	PM ₁₄	PM ₂₁	PM ₂₂	PM ₂₃	PM ₂₄	
Target points	12.6	9.5	25.2	14.7	3.2	2.4	6.4	3.6	77.5
Mean	13.7	10.3	27.4	17.1	3.4	2.6	6.9	4.3	85.6
STDEV	0.4	0.3	0.9	0.5	0.1	0.1	0.2	0.1	2.5
Min	13.3	10.0	26.6	16.6	3.3	2.5	6.5	4.1	83.2
Max	14.3	10.8	28.7	17.9	3.6	2.7	7.2	4.5	89.7
25 th percentile	13.3	10.0	26.6	16.6	3.4	2.5	6.7	4.2	83.6
Median	13.5	10.1	26.9	16.8	3.4	2.6	6.9	4.3	84.2
75 th percentile	14.1	10.6	28.2	17.6	3.5	2.6	6.9	4.3	87.8
Boxplot LB whisker	12.1	9.1	24.3	15.2	3.1	2.3	6.2	3.9	76.9
Boxplot UB whisker	15.3	11.4	30.5	19.1	3.7	2.7	7.3	4.6	94.1

Findings: The transportation project delivery process involves four key phases: planning, design, construction, and operations management. Mastery of these stages requires a solid grounding in the core concepts that underpin each. The two graduate-level courses provide essential foundational knowledge in these subject areas, making them critical for students specializing in transportation engineering.

- **Course-Specific Student Learning Performance:** Overall, students in each class have earned average scores that exceed the target performance thresholds. The standard deviations of these averages are consistently low, ranging from approximately 2.1% to 2.5%, indicating a relatively uniform level of achievement across students. Key details are highlighted below:
 - For both courses, the lowest and highest student scores fall within the lower and upper whiskers of the boxplots, indicating no student performed exceptionally poorly or extraordinarily well relative to their peers.
 - When comparing the two classes, the mean score for CAE 568 – Transportation Asset Management (85.6/100) is slightly higher than that of CAE 544 – Urban Transportation Planning (84/100).
 - Analyzing the relationship between the sample mean and median reveals slight skewness in both distributions. For CAE 544, the mean is marginally less than the median, indicating a slight left skew, with more than 5 out of 11 students scoring above the course average. For CAE 568, the mean exceeds the median, indicating a slight right skew, with more than 3 out of 7 students scoring below the course average.
- **Subject-Area Performance of Student Learning:** As shown in Table 3, student performance across the six subject areas covered in the two courses can be grouped into a single overall performance category:
 - Students exceeded expectations in both acquiring and applying fundamental knowledge across all six subject areas covered in the two courses. In CAE 544 – Urban Transportation Planning, students demonstrated strong mastery in travel demand forecasting and demand/supply equilibrium. In CAE 568 – Transportation Asset Management, students also performed above target levels in transportation data collection, processing, and database management; transportation system performance modeling; physical facility management; and system usage management.
- Student learning effectiveness was further evaluated using specific performance measures, as summarized in Table 4. For both CAE 544 and CAE 568, performance measures PM11, PM12, PM13, and PM14 were used to assess Goal 1 (knowledge acquisition), while PM21, PM22, PM23, and PM24 were used to assess Goal 2 (knowledge application). Together, these metrics provide a clearer picture of how well students achieved the intended learning outcomes.

Discussions: Students in both classes performed well overall. Across the two learning goals spanning six subject areas, no deficiencies were observed in the application of knowledge (Goal 2) for travel demand forecasting, which is the one subject area common to both courses.

Improvement Plan: No action is required at this time. However, the instructor for CAE 544 and CAE 568 (Zongzhi Li, PhD, Full Professor) will continue to monitor student learning outcomes and implement improvement measures as needed in the future.